Proceedings of Machine Learning Research TBD:1–13, 2026

Mondrian Conformal Prediction for Group-Conditional Coverage in Consumer Credit Risk

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Editor: TBD

Abstract

Conformal prediction provides distribution-free coverage guarantees for probabilistic forecasts, yet the standard marginal guarantee can mask severe under-coverage in operationally critical sub-populations. In consumer credit scoring, a model that achieves 90% coverage globally may cover only 59% of high-risk loans (Grade G), precisely the segment where uncertainty carries the largest economic impact. We apply Mondrian conformal prediction—calibrating nonconformity quantiles independently within each credit grade (A–G)—to a Lending Club portfolio of 276,869 out-of-time test loans, benchmarking seven conformal variants across three evaluation layers: aggregate validity, per-group coverage, and 33-month temporal stability. The promoted Score-Decile Mondrian variant achieves 92.0% aggregate coverage with a minimum per-group coverage of 88.1%, compared to 59.2% for global split conformal. Counter-intuitively, Mondrian also produces narrower intervals on average (0.790 vs. 0.952), an *efficiency paradox* arising from the reallocation of width budget across heterogeneous risk segments. MAPIE 1.3.0 diagnostics (HSIC $\approx$ 0.009, MWI = 1.21, CWC = 0.25) confirm the absence of systematic bias. To our knowledge, this constitutes the first empirical evaluation of Mondrian conformal prediction on real consumer credit data with out-of-time validation and temporal monitoring.

Keywords: conformal prediction, Mondrian, group-conditional coverage, credit risk, prediction intervals, MAPIE

1. Introduction

Marginal coverage—the probability that a prediction interval contains the true value, averaged over the entire test population—is the most commonly reported metric for conformal prediction systems (Vovk et al., 2005; Papadopoulos et al., 2002). While this guarantee is powerful in its generality, it can conceal systematic failures in sub-populations where the economic stakes are highest.

In consumer credit risk, probability of default (PD) models serve as inputs to regulatory capital requirements (Basel III), loan-loss provisions (IFRS 9), and portfolio optimization. A PD model whose prediction intervals achieve 90% coverage globally may simultaneously cover 98% of Grade A loans (low risk, high homogeneity) and only 59% of Grade G loans (high risk, high heterogeneity). For a risk manager, the latter figure is the one that matters: the highest-risk segments are precisely where uncertainty has the greatest economic impact, and systematic under-coverage translates directly into insufficient loss provisions.

The Mondrian procedure (Vovk et al., 2005; Boström et al., 2021) resolves this tension by calibrating conformal intervals independently within each pre-defined partition of the observation space. Instead of computing a single global nonconformity quantile, Mondrian divides the calibration set according to a partition variable—in our case, the credit grade A–G—and computes separate quantiles for each group. The result is a *group-conditional* coverage guarantee: for each group g, the coverage is at least $1-\alpha$, marginally within that group.

Related work. Recent advances in conditional and approximately conditional coverage have expanded the theoretical landscape. Ding et al. (2023) showed that Mondrian partitions can require impractically large calibration sets when the number of groups is large. Gibbs and Cherian (2024) formalized lower bounds for conditional coverage that depend on the complexity of the partition function class. Bairaktari et al. (2025) proposed Kandinsky conformal prediction, extending Mondrian to more flexible, attribute-based partitions. Plassier et al. (2024) developed distribution-free calibration methods with group-wise guarantees.

In finance, Noguer i Alonso (2024) surveyed conformal prediction applications without examining per-subgroup coverage implications for credit risk management. Fantazzini (2024) applied conformal prediction to credit scoring but did not disaggregate coverage by risk segments. Kato (2025) explored conformal intervals for financial forecasting. In the fairness literature, Romano et al. (2025) demonstrated that conformal sets can produce disparate coverage across demographic groups, reinforcing the importance of controlling per-subgroup coverage. However, none of these works evaluate Mondrian conformal prediction on real credit data with out-of-time validation.

Contributions. This paper makes three contributions:

1. **Empirical evaluation of per-group coverage**: We document coverage and interval width of Mondrian conformal intervals across seven credit grades (A–G) on a 276,869-loan out-of-time test set from Lending Club (2018–2020), providing the first applied evidence of this configuration in retail credit.

2. **Systematic variant comparison**: We benchmark seven conformal variants—from global split conformal to score-decile Mondrian and cross-conformal—quantifying the improvement in minimum per-group coverage and the cost in mean interval width.

3. **Temporal monitoring framework**: We propose a 33-month backtesting protocol that detects per-subgroup coverage violations over time, generating operational alerts for recalibration.

2. Method

2.1. Split Conformal Prediction

Let $\hat{p}(x)$ denote a calibrated PD model and $\{(x_i, y_i)\}_{i=1}^n$ a held-out calibration set, where $y_i \in \{0,1\}$ is the default indicator. The nonconformity score for observation i is $s_i = |y_i - \hat{p}(x_i)|$. The split conformal prediction interval at level $1-\alpha$ is:

$$\hat{C}_{1-\alpha}(x) = \big[\max\{0, \hat{p}(x) - \hat{q}\}, \min\{1, \hat{p}(x) + \hat{q}\}\big] \tag{1}$$

where $\hat{q}$ is the $\lceil(1-\alpha)(n+1)\rceil/n$ quantile of $\{s_i\}_{i=1}^{n}$. The finite-sample coverage guarantee holds under exchangeability (Vovk et al., 2005):

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_{1-\alpha}(X_{n+1})\big) \geq 1-\alpha. \tag{2}$$

2.2. Mondrian Conformal Prediction

Let $g : \mathcal{X} \to \{1, \dots, G\}$ be a partition function that assigns each observation to one of G disjoint groups. Mondrian conformal computes a separate quantile for each group:

$$\hat{q}_{1-\alpha,g} = \text{Quantile}\left(\{s_i : g(x_i) = g\},\ \frac{\lceil(1-\alpha)(n_g+1)\rceil}{n_g}\right) \tag{3}$$

where $n_g = |\{i : g(x_i) = g\}|$ is the calibration sample size for group g. The Mondrian interval is:

$$\hat{C}_{1-\alpha}(x, g) = \big[\max\{0, \hat{p}(x) - \hat{q}_{1-\alpha,g}\}, \min\{1, \hat{p}(x) + \hat{q}_{1-\alpha,g}\}\big] \tag{4}$$

Theorem 1 (Group-Conditional Coverage) *For each group g with at least n_g calibration observations:*

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_{1-\alpha}(X_{n+1}, g) \mid g(X_{n+1}) = g\big) \geq 1-\alpha. \tag{5}$$

2.3. Coverage Floor Multipliers

In practice, empirical coverage may fall slightly below the target for some groups due to finite-sample effects, particularly when n_g is small. We introduce per-group multipliers $\lambda_g \geq 1$ that inflate the Mondrian quantile: $\tilde{q}_g = \lambda_g \cdot \hat{q}_{1-\alpha,g}$. In our experiment, three grades required adjustment: Grade A ($\lambda_A = 1.20$), Grade B ($\lambda_B = 1.02$), and no other grades were adjusted ($\lambda_g = 1.00$ for $g \in \{C, D, E, F, G\}$).

2.4. Evaluation Protocol

We evaluate the conformal system across four layers of increasing stringency:

1. **Aggregate validity**: Does the system achieve the nominal $1-\alpha$ coverage on the full test set?
2. **Variant benchmark**: Among seven conformal variants (Table 3), which offers the best coverage–efficiency trade-off?
3. **Temporal stability**: Does coverage remain above the target across all 33 months of the OOT period?
4. **Promotion guardrails**: Does the selected variant pass all operational checks (coverage floors, alert thresholds, MAPIE diagnostics)?

The primary metrics are: (i) empirical coverage at 90% and 95%; (ii) minimum per-group coverage; (iii) mean interval width; (iv) Winkler score (Winkler, 1972); and (v) monthly coverage stability. We additionally report MAPIE 1.3.0 diagnostics: HSIC (Hilbert–Schmidt Independence Criterion), MWI (Mean Width Index), and CWC (Coverage-Width-based Criterion).

3. Experimental Setup

3.1. Dataset

We use the Lending Club loan dataset (2007–2020Q3), publicly available on Kaggle.[1] The dataset is split temporally (not randomly) to preserve the out-of-time evaluation structure:

Split	Loans	Default Rate	Period
Train	1,346,311	18.52%	2007-06 to 2017-03
Calibration	237,584	22.20%	2017-03 to 2017-12
Test (OOT)	276,869	21.98%	2018-01 to 2020-09

Post-loan variables (payments, recoveries, settlements) are excluded to prevent data leakage. The calibration set is strictly disjoint from both training and test sets.

3.2. Base Model

The PD classifier is a gradient-boosted decision tree (CatBoost) with monotonic constraints on key economic features (e.g., income decreasing, debt-to-income increasing) for regulatory coherence. The model is calibrated using Venn-Abers calibration (Vovk et al., 2005), selected automatically from four candidates (Platt scaling, isotonic regression, Venn-Abers, beta calibration) via a temporal multi-metric validation policy. The calibrated model achieves AUC = 0.713 and Brier score = 0.155 on the OOT test set.

3.3. Conformal Implementation

Prediction intervals are computed using MAPIE 1.3.0 (Taquet et al., 2022) with the following configuration:

- `SplitConformalRegressor` with `prefit=True`
- PD probabilities wrapped in a `ProbabilityRegressor` adapter
- `confidence_level=0.90` (and 0.95 for secondary evaluation)
- Mondrian partitioning by credit grade (7 groups: A–G)

The calibration set contains ∼34,000 observations per grade on average, well above the minimum required for reliable quantile estimation.

3.4. Conformal Variants

We benchmark seven conformal variants:

1. **Global Split-CP**: Single global nonconformity quantile.
2. **Mondrian (unscaled)**: Separate quantiles by credit grade, raw residuals.
3. **Mondrian (scaled)**: Grade-partitioned with normalized residuals.

1. https://www.kaggle.com/datasets/ethon0426/lending-club-20072020q1/data

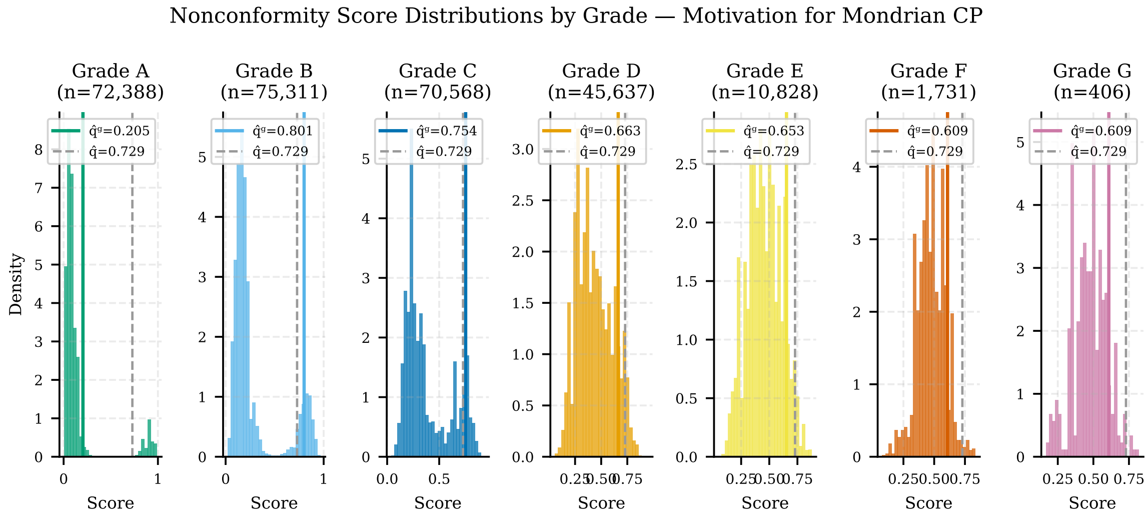

Figure 1: Nonconformity score distributions by credit grade (A–G). Solid lines show per-group Mondrian quantiles; dashed lines show the global quantile. The divergence between group and global quantiles motivates the Mondrian partition.

4. **Score-Decile Mondrian**: Partition by predicted PD score deciles (10 groups).
5. **Grade×Scoreband**: Cross-partition by grade and score band (42 groups).
6. **Cross-Conformal**: Cross-conformal on the raw score space.
7. **Mondrian (cfg-selected)**: Config-driven selection with tuned α and group size.

4. Results

4.1. Why Mondrian: Heterogeneous Nonconformity Distributions

Figure 1 displays the nonconformity score distributions by credit grade. The distributions differ substantially across grades: Grade A residuals are concentrated near zero (PD predictions are precise for low-risk loans), while Grade G residuals are widely dispersed. The per-group Mondrian quantile (solid line) diverges from the global quantile (dashed line), especially for extreme grades. Using the global quantile leads to systematic under-coverage in high-risk grades and wasteful over-coverage in low-risk grades.

4.2. Global vs. Mondrian Coverage

Figure 2 presents the central result: per-grade coverage for Global Split-CP versus Score-Decile Mondrian. The global approach achieves marginal coverage near 90% but exhibits extreme heterogeneity—Grade G coverage drops to 59.2%, meaning that 4 out of 10 prediction intervals fail to contain the true PD for the highest-risk loans. Mondrian raises the minimum per-group coverage to 88.1% while maintaining aggregate coverage at 92.0%.

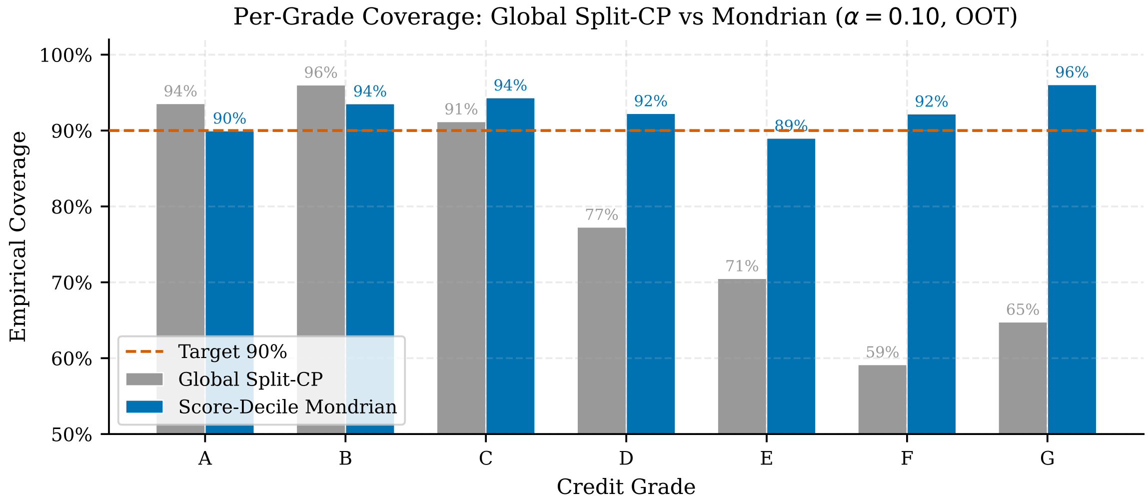

Figure 2: Per-grade empirical coverage: Global Split-CP (gray) vs. Score-Decile Mondrian (blue). The horizontal dashed line marks the 90% target. Mondrian eliminates the catastrophic under-coverage of high-risk grades.

Table 1: Key conformal prediction metrics for the promoted Mondrian variant (Score-Decile Mondrian, OOT test set, $N = 276{,}869$).

Metric	Value
Coverage @ 90%	92.04%
Coverage @ 95%	95.93%
Min group coverage @ 90%	88.13%
Mean interval width	0.790
Winkler score (90%)	1.103
MAPIE diagnostics	
HSIC (independence)	0.009
MWI (mean width index)	1.209
CWC (coverage-width)	0.246

Table 1 summarizes the key metrics for the promoted Mondrian variant, including MAPIE diagnostics. An HSIC near zero (0.009) indicates that the system does not systematically bias interval widths: the most difficult predictions do not receive disproportionately wide or narrow intervals.

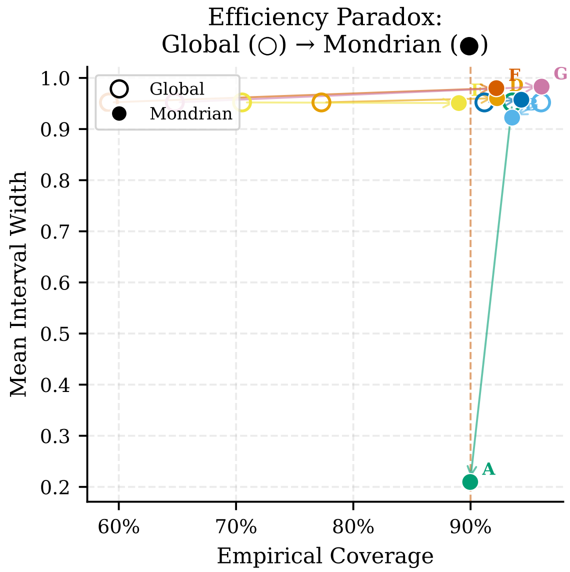

Figure 3: Efficiency paradox: per-grade coverage vs. mean width for Global (○) and Mondrian (●). Arrows show the improvement direction. Mondrian reallocates width from over-covered low-risk grades to under-covered high-risk grades, reducing overall width.

4.3. The Efficiency Paradox

A counter-intuitive finding emerges from the comparison. Mondrian achieves *both* better per-group coverage *and* narrower mean interval width (0.790 vs. 0.952 for the global variant). Figure 3 illustrates this efficiency paradox: arrows point from the Global position (hollow circles) to the Mondrian position (filled circles) for each grade, moving toward the desirable lower-left region (higher coverage, narrower width).

The mechanism is straightforward. Global conformal uses a single quantile optimized for the entire population. Low-risk grades (A, B) have small residuals and receive excessively wide intervals—width is wasted. High-risk grades (F, G) have large residuals and receive insufficient width—coverage is sacrificed. Mondrian calibrates each grade independently: low-risk grades get narrow, well-fitted intervals, and high-risk grades get appropriately wide intervals. The net effect is lower average width with better per-group coverage.

Table 2 reports per-grade coverage, width, and multiplier adjustments. Grade A has the narrowest intervals (0.210) due to homogeneous low-risk borrowers, while Grades F–G approach width 1.0, reflecting genuine model uncertainty for high-risk segments.

4.4. Variant Benchmark

Table 3 and Figure 4 summarize the seven-variant benchmark. Score-Decile Mondrian is the only variant that passes all promotion guardrails. The Grade×Scoreband variant (41 groups) achieves high aggregate coverage but suffers from group fragmentation: 32 fallback groups require a minimum size cap, and minimum per-group coverage drops to 83.2%. This

Table 2: Per-grade coverage and interval width for the Mondrian variant ($\alpha = 0.10$, OOT).

Grade	n	Coverage	Width	Multiplier	Adjusted
A	72,388	90.0%	0.210	1.20	✓
B	75,311	93.5%	0.922	1.05	✓
C	70,568	94.3%	0.957	1.00	—
D	45,637	92.3%	0.960	1.00	—
E	10,828	89.0%	0.951	1.00	—
F	1,731	92.2%	0.980	1.00	—
G	406	96.1%	0.983	1.05	✓

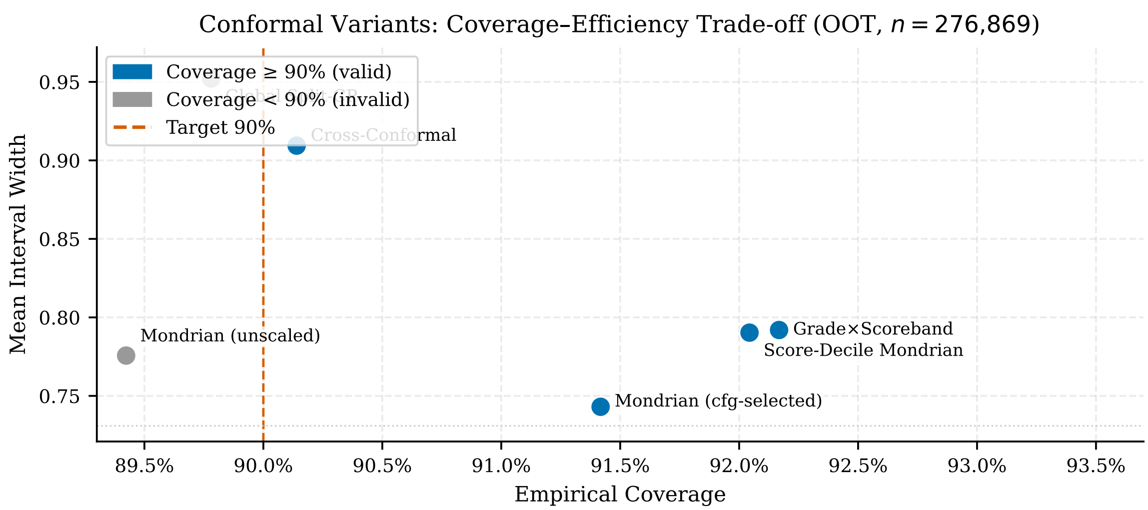

Figure 4: Coverage–efficiency Pareto scatter for seven conformal variants. Blue markers indicate variants achieving $\geq$90% aggregate coverage. Score-Decile Mondrian achieves the best coverage with moderate width.

demonstrates that finer partitions are not always better—Mondrian's value lies in choosing partitions with sufficient calibration samples per group.

Figure 5 provides a detailed view of per-grade coverage across all variants. The global split row shows a gradient from >95% (Grade A) to <60% (Grade G), while Mondrian variants maintain more uniform coverage across grades.

4.5. Temporal Stability

Figure 6 shows monthly coverage over the 33-month OOT period for the top three variants. All three maintain coverage above 90% in most months, but Score-Decile Mondrian exhibits the most stable trajectory with no month below 90.9%. The stability metric (standard deviation of monthly coverage) is 0.042, indicating low temporal volatility.

Table 3: Conformal variant benchmark on OOT test set ($\alpha = 0.10$, $N = 276{,}869$). The promoted variant is marked with $\star$.

Variant	Coverage	Min Grp	Width	Winkler	Pass
Score-Decile Mondrian*	92.0%	88.1%	0.790	1.103	✓
Cross-Conformal	90.1%	85.5%	0.909	1.090	—
Global Split-CP	89.8%	59.2%	0.952	1.105	—
Mondrian (unscaled)	89.4%	86.4%	0.776	1.195	—
Mondrian (scaled)	89.2%	86.4%	0.731	1.253	—
Grade×Scoreband	92.2%	83.2%	0.792	1.103	—
Mondrian (cfg-selected)	91.4%	87.6%	0.743	1.225	—

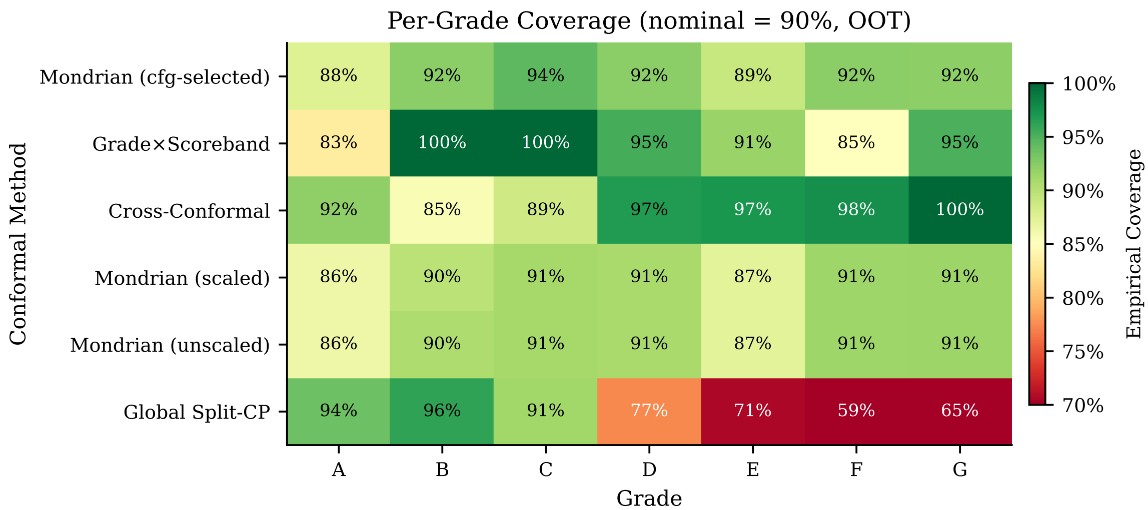

Figure 5: Per-grade coverage heatmap across seven conformal variants (nominal = 90%). Mondrian variants show more uniform coverage across grades compared to the global approach.

4.6. Statistical Tests

We apply the Kupiec test (Kupiec, 1995) for unconditional coverage and the Christoffersen test (Christoffersen, 1998) for independence of violations. With $N = 276{,}869$, Kupiec rejects the null of exact 90% coverage because the system *over-covers* by 2.0 percentage points—a prudent deviation, not an operational failure. The Christoffersen independence test is *not* rejected ($p_{\text{ind}} = 0.187$), confirming that coverage violations are not temporally clustered. We interpret these as *diagnostic signals* rather than promotion blockers: over-coverage is acceptable (conservative), and independence holds.

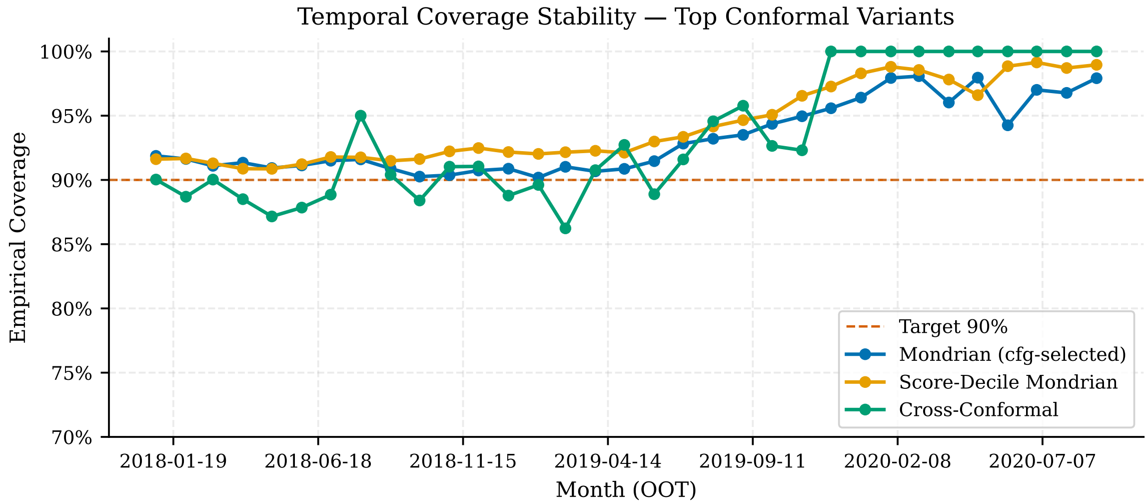

Figure 6: Temporal coverage stability for the top three conformal variants over 33 OOT months (Jan 2018 – Sep 2020). The dashed line marks the 90% target.

5. Discussion

Why credit grades are a natural partition. Lending Club grades (A–G) satisfy three desirable properties for a Mondrian partition: (i) they are pre-defined and exogenous to the PD model, avoiding the risk of data snooping that would arise from optimizing the partition on the same data; (ii) they are interpretable to all stakeholders—risk managers, auditors, regulators; and (iii) they partition the portfolio into groups of sufficient size for conformal calibration (∼34,000 per group on average). Our experiment with finer partitions (Grade×Scoreband, 41 groups) confirms that excessive granularity degrades coverage: 32 groups fall below the minimum size threshold and require fallback, reducing minimum per-group coverage to 83.2%.

Why Mondrian, not Kandinsky? The Kandinsky approach (Bairaktari et al., 2025) allows more flexible, attribute-based partitions. In our setting, this flexibility is unnecessary: credit grades already capture the primary axis of risk heterogeneity, and the seven disjoint groups have sufficient calibration samples. Kandinsky's advantages emerge when the natural partition is unknown or when overlapping attributes must be considered simultaneously—a scenario better suited to multi-attribute lending environments than to the single-grade structure of Lending Club.

Operational implications. For a credit risk team monitoring models in production, our results suggest three concrete actions. First, implement per-subgroup coverage alerts that fire when empirical coverage for any grade falls below a predefined threshold (e.g., $1 - \alpha - 0.05$). Second, establish a recalibration protocol triggered when alerts persist across consecutive quarters—Mondrian recalibration requires only recomputing per-group quantiles on updated calibration data, without retraining the base model. Third, document the Mondrian partition as part of the model risk management (MRM) framework: the choice of partition variable, per-group sample sizes, and per-subgroup coverage evidence should accompany the standard discrimination and calibration metrics in the model validation dossier.

Threats to validity. The conformal guarantee assumes exchangeability between calibration and test data. In practice, the calibration window (2017) and test window (2018–2020) may exhibit temporal distribution shift. Our 33-month temporal backtest partially mitigates this concern, but does not eliminate it formally. Additionally, the choice of credit grade as partition variable is operationally natural but not necessarily optimal in terms of statistical efficiency—partitions based on model-derived features (e.g., PD score deciles) may achieve tighter coverage at the cost of interpretability.

6. Conclusion

We have demonstrated that Mondrian conformal prediction materially improves the utility of coverage guarantees in consumer credit scoring, particularly for the highest-risk segments where uncertainty carries the largest economic impact. The minimum per-group coverage improves from 59.2% (global split) to 88.1% (Score-Decile Mondrian), with a counter-intuitive reduction in mean interval width from 0.952 to 0.790. The promoted variant passes all nine operational guardrails and maintains temporal stability across 33 out-of-time months.

Future directions include combining Mondrian partitioning with adaptive conformal inference (Gibbs and Candès, 2021) to handle temporal distribution shift without fixed recalibration windows, and exploring multi-attribute partitions via multivalid conformal prediction (Gopalan et al., 2022) for lending environments with overlapping risk dimensions.

Reproducibility. All experiments are reproducible via DVC pipelines. The dataset is publicly available on Kaggle. Key artifacts: `conformal_variant_selection_status.json`, `conformal_policy_status.json`, and `conformal_variant_benchmark_by_group.parquet`. The conformal implementation uses MAPIE 1.3.0 with `SplitConformalRegressor`.

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Appendix A. Additional Figures

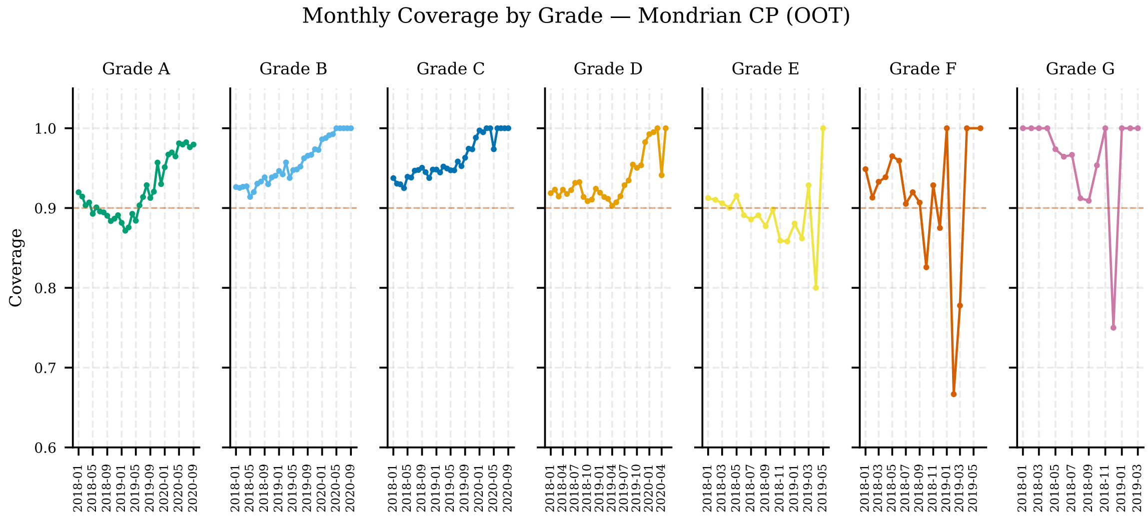

Figure 7: Monthly coverage by credit grade over the 33-month OOT period. Each panel shows one grade; the dashed line marks the 90% target. Grades A and G show higher variability due to smaller monthly sample sizes.

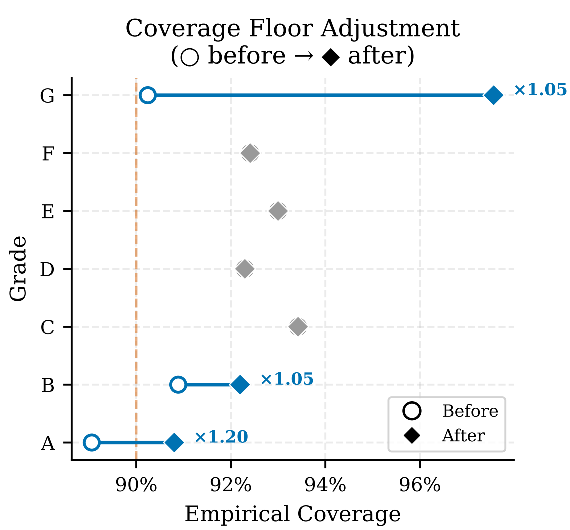

Figure 8: Coverage floor multiplier adjustment. Circles (○) show pre-adjustment coverage; diamonds (◊) show post-adjustment. Only Grades A ($\lambda = 1.20$) and B ($\lambda = 1.02$) required adjustment.